

Exogenous and Endogenous Risk Matrices in Downside Portfolio Optimization

Endogenous Markowitz Semivariance, Estrada’s Exogenous Heuristic, and
Viole–Nawrocki/NNS Directional Co-Partial Moment Matrices

Fred Viole

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Abstract

This note formalizes the distinction between **endogenous** and **exogenous** risk matrices in portfolio optimization when risk is measured by semivariance, lower partial moments, upper partial moments, or co-partial moment analogs rather than classical variance. The distinction is structural: an **endogenous** matrix depends on the portfolio weights being optimized, whereas an **exogenous** matrix is computed entirely from the asset-return sample and a fixed benchmark before optimization begins.

This distinction explains why traditional mean-semivariance optimization is computationally difficult, why Estrada’s heuristic restores quadratic-program tractability, and why the Viole–Nawrocki/NNS co-partial-moment framework provides a broader directional solution using benchmark-relative pairwise dependence matrices.

The central conclusion is precise: Viole–Nawrocki do **not** make the original Markowitz portfolio-semivariance matrix exogenous. Rather, they replace that endogenous object with exogenous directional primitives. The endogeneity problem is eliminated by design.

1. Setup and Notation

Let there be N assets and T observations. Let

$$R_{i,t}$$

denote the return of asset i at time t , and let

$$\mathbf{x} = (x_1, \dots, x_N)^\top$$

denote the portfolio weight vector, subject to constraints such as

$$\sum_{i=1}^N x_i = 1, \quad x_i \geq 0.$$

The portfolio return at time t is

$$R_{p,t}(\mathbf{x}) = \sum_{i=1}^N x_i R_{i,t}.$$

Let B or h denote a fixed benchmark or target return.

Classical mean-variance optimization uses the covariance matrix

$$\Sigma = [\sigma_{ij}]_{i,j=1}^N,$$

where

$$\sigma_{ij} = \frac{1}{T} \sum_{t=1}^T (R_{i,t} - \mu_i)(R_{j,t} - \mu_j).$$

Once estimated from the return sample, Σ is fixed. Portfolio variance is then

$$\sigma_p^2(\mathbf{x}) = \mathbf{x}^\top \Sigma \mathbf{x}.$$

The key advantage is that the risk matrix is an input to the optimizer, not an output of the optimizer.

2. Definition: Exogenous vs. Endogenous Risk Matrices

A risk matrix $\mathbf{M}$ is **exogenous** if it can be computed before optimization from the observed asset returns and fixed parameters only:

$$\mathbf{M} = \mathbf{M}(\{R_{i,t}\}_{i,t}, B).$$

Equivalently,

$$\frac{\partial \mathbf{M}}{\partial x_k} = 0 \quad \text{for all } k.$$

A risk matrix is **endogenous** if its elements depend on the portfolio weights:

$$\mathbf{M} = \mathbf{M}(\mathbf{x}),$$

so that generally

$$\frac{\partial \mathbf{M}(\mathbf{x})}{\partial x_k} \neq 0 \quad \text{for some } k.$$

This distinction matters because a quadratic portfolio risk objective of the form

$$\mathbf{x}^\top \mathbf{M} \mathbf{x}$$

is a standard quadratic form only when $\mathbf{M}$ is fixed. If $\mathbf{M}$ itself changes with $\mathbf{x}$, then the objective is no longer an ordinary quadratic program in the usual mean-variance sense.

Estrada frames this directly: mean-semivariance optimization is difficult because, unlike the exogenous covariance matrix used in mean-variance optimization, the semicovariance matrix is endogenous. His contribution is to propose a heuristic that yields a symmetric and exogenous semicovariance matrix usable with the same closed-form machinery as mean-variance optimization.

3. Traditional Portfolio Semivariance and Endogeneity

The semivariance of a portfolio relative to benchmark B is

$$\Sigma_{pB}^2(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^T [\min(R_{p,t}(\mathbf{x}) - B, 0)]^2.$$

Equivalently, the downside event is determined by the condition

$$R_{p,t}(\mathbf{x}) < B.$$

A traditional portfolio-conditioned cosemivariance matrix may be written schematically as

$$S_{ijB}(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\{R_{p,t}(\mathbf{x}) < B\}} (R_{i,t} - B)(R_{j,t} - B).$$

The decisive feature is the indicator

$$\mathbf{1}_{\{R_{p,t}(\mathbf{x}) < B\}}.$$

Because

$$R_{p,t}(\mathbf{x}) = \sum_{k=1}^N x_k R_{k,t},$$

the set of time indices classified as “downside” depends on the portfolio weights. A change in $\mathbf{x}$ can change which observations enter the downside region. Therefore

$$S_{ijB}(\mathbf{x})$$

is endogenous.

Proposition 1: Traditional Cosemivariance Is Endogenous

Define

$$\mathcal{D}(\mathbf{x}) = \{t : R_{p,t}(\mathbf{x}) < B\}.$$

Then

$$S_{ijB}(\mathbf{x}) = \frac{1}{T} \sum_{t \in \mathcal{D}(\mathbf{x})} (R_{i,t} - B)(R_{j,t} - B).$$

Since $\mathcal{D}(\mathbf{x})$ depends on $\mathbf{x}$, the summation domain itself depends on the optimizer's decision variables. Thus $S_{ijB}(\mathbf{x})$ is not known before the optimization problem is solved. It is endogenous.

4. Consequence for Optimization

A standard mean-variance problem has the form

$$\min_{\mathbf{x}} \quad \mathbf{x}^\top \Sigma \mathbf{x}$$

subject to constraints such as

$$\mathbf{x}^\top \boldsymbol{\mu} \geq \mu^*, \quad \mathbf{1}^\top \mathbf{x} = 1.$$

Because Σ is fixed, this is a quadratic program.

The analogous mean-semivariance problem is

$$\min_{\mathbf{x}} \quad \Sigma_{pB}^2(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^T [\min(R_{p,t}(\mathbf{x}) - B, 0)]^2.$$

If one attempts to express this as

$$\mathbf{x}^\top S_B(\mathbf{x}) \mathbf{x},$$

the matrix $S_B(\mathbf{x})$ still depends on the unknown $\mathbf{x}$. Hence the optimizer must repeatedly recompute downside states as the portfolio weights change. This explains why the exact traditional formulation typically requires numerical algorithms rather than the ordinary closed-form or quadratic-program machinery of mean-variance optimization.

5. Estrada's Exogenous Semicovariance Heuristic

Estrada's solution is to define downside at the **asset level**, not the portfolio level. For a fixed benchmark B , define

$$\Sigma_{ijB} = \frac{1}{T} \sum_{t=1}^T \min(R_{i,t} - B, 0) \min(R_{j,t} - B, 0).$$

Here the downside transformation is applied separately to each marginal asset return. Define

$$d_{i,t}^- = \min(R_{i,t} - B, 0).$$

Then

$$\Sigma_{ijB} = \frac{1}{T} \sum_{t=1}^T d_{i,t}^- d_{j,t}^-.$$

This matrix is independent of $\mathbf{x}$. Therefore

$$\frac{\partial \Sigma_{ijB}}{\partial x_k} = 0 \quad \text{for all } i, j, k.$$

Thus Σ_B is exogenous.

It is also symmetric, because

$$\Sigma_{ijB} = \frac{1}{T} \sum_{t=1}^T d_{i,t}^- d_{j,t}^- = \frac{1}{T} \sum_{t=1}^T d_{j,t}^- d_{i,t}^- = \Sigma_{jiB}.$$

Estrada's approximation to portfolio semivariance is then

$$\Sigma_{pB}^2 \approx \sum_{i=1}^N \sum_{j=1}^N x_i x_j \Sigma_{ijB} = \mathbf{x}^\top \Sigma_B \mathbf{x}.$$

This restores the mean-variance form. The matrix is computed once, before optimization, and then passed into the quadratic optimizer as a fixed input.¹

However, this is still a heuristic approximation to exact portfolio semivariance. Estrada solves the computational issue by changing the matrix definition, not by making the original portfolio-conditioned matrix exogenous.

¹The benchmark B (equivalently h) is fixed by the investor's target return before optimization begins, playing the same structural role as B in Estrada's original formulation; sensitivity to the benchmark choice is therefore a user input, not a source of endogeneity.

6. The Viole–Nawrocki/NNS Extension: Co-Partial Moment Matrices

The Viole–Nawrocki/NNS framework generalizes the exogenous-matrix idea using **directional partial moments** and **co-partial moments**.

The univariate lower and upper partial moments are

$$LPM(n, h; X) = \frac{1}{T} \sum_{t=1}^T \max(h - X_t, 0)^n,$$

and

$$UPM(n, h; X) = \frac{1}{T} \sum_{t=1}^T \max(X_t - h, 0)^n.$$

For a shared benchmark h , define the lower directional deviation

$$\ell_{i,t}^{(n)} = \max(h - X_{i,t}, 0)^n,$$

and the upper directional deviation

$$u_{i,t}^{(n)} = \max(X_{i,t} - h, 0)^n.$$

Then the concordant lower co-partial moment is

$$\text{CoLPM}_{ij}^{(n)}(h) = \frac{1}{T} \sum_{t=1}^T \ell_{i,t}^{(n)} \ell_{j,t}^{(n)}.$$

The concordant upper co-partial moment is

$$\text{CoUPM}_{ij}^{(n)}(h) = \frac{1}{T} \sum_{t=1}^T u_{i,t}^{(n)} u_{j,t}^{(n)}.$$

These are pairwise, benchmark-relative, directional dependence measures. They do not require the portfolio return $R_{p,t}(\mathbf{x})$ and do not require knowing which portfolio observations fall below the benchmark. Each element is computed only from the pair (X_i, X_j) , the sample index t , the benchmark h , and the degree n .

Therefore these matrices are exogenous by construction.

7. Gram-Matrix Structure and Positive Semidefiniteness

Let

$$\mathbf{L} = \begin{bmatrix} \ell_{1,1}^{(n)} & \ell_{2,1}^{(n)} & \cdots & \ell_{N,1}^{(n)} \\ \ell_{1,2}^{(n)} & \ell_{2,2}^{(n)} & \cdots & \ell_{N,2}^{(n)} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{1,T}^{(n)} & \ell_{2,T}^{(n)} & \cdots & \ell_{N,T}^{(n)} \end{bmatrix}.$$

Then the lower co-partial moment matrix can be written as

$$\text{CoLPM}^{(n)}(h) = \frac{1}{T} \mathbf{L}^\top \mathbf{L}.$$

For any vector $\mathbf{x}$,

$$\mathbf{x}^\top \text{CoLPM}^{(n)}(h) \mathbf{x} = \frac{1}{T} \mathbf{x}^\top \mathbf{L}^\top \mathbf{L} \mathbf{x} = \frac{1}{T} \|\mathbf{L} \mathbf{x}\|^2 \geq 0.$$

Thus

$$\text{CoLPM}^{(n)}(h)$$

is positive semidefinite.

Similarly, if $\mathbf{U}$ is the matrix of upper directional deviations, then

$$\text{CoUPM}^{(n)}(h) = \frac{1}{T} \mathbf{U}^\top \mathbf{U}$$

and

$$\mathbf{x}^\top \text{CoUPM}^{(n)}(h) \mathbf{x} = \frac{1}{T} \|\mathbf{U} \mathbf{x}\|^2 \geq 0.$$

Thus

$$\text{CoUPM}^{(n)}(h)$$

is also positive semidefinite.

A sum of positive semidefinite matrices is always positive semidefinite. It is positive definite when the combined directional-deviation design spans all nonzero portfolio directions, i.e., when no nonzero $\mathbf{x}$ lies in the joint null space of the component matrices.

For example, positive definiteness can fail if an asset has zero directional deviation in every period relative to the benchmark, or if one asset's directional-deviation vector is an exact linear combination of others. In those cases, the matrix remains positive semidefinite but may be singular.

8. Relationship to Classical Variance and Covariance

The NNS co-partial-moment framework does not merely provide an alternative to semivariance. It also decomposes the classical covariance matrix itself into directional building blocks.

Let X_i and X_j be centered relative to their benchmarks, typically their means:

$$h_i = \mu_i, \quad h_j = \mu_j.$$

Define the degree-1 directional deviations

$$L_i = \max(h_i - X_i, 0), \quad U_i = \max(X_i - h_i, 0),$$

and similarly for asset j . Then the signed deviation can be written as

$$X_i - h_i = U_i - L_i.$$

Therefore the covariance between X_i and X_j is

$$\text{Cov}(X_i, X_j) = E[(X_i - h_i)(X_j - h_j)].$$

Substituting the directional decomposition gives

$$\text{Cov}(X_i, X_j) = E[(U_i - L_i)(U_j - L_j)].$$

Expanding,

$$\text{Cov}(X_i, X_j) = E[U_i U_j] + E[L_i L_j] - E[L_i U_j] - E[U_i L_j].$$

Thus, in co-partial-moment notation,

$$\boxed{\Sigma = \text{CoLPM}^{(1)} + \text{CoUPM}^{(1)} - \text{CoDLPM}^{(1)} - \text{CoDUPM}^{(1)}}$$

where:

- $\text{CoLPM}^{(1)}$ captures concordant lower-side co-movement;
- $\text{CoUPM}^{(1)}$ captures concordant upper-side co-movement;
- $\text{CoDLPM}^{(1)}$ captures the case where asset i is below target while asset j is above target;
- $\text{CoDUPM}^{(1)}$ captures the case where asset i is above target while asset j is below target.

This identity shows that classical covariance is not outside the directional framework. It is a signed linear combination of exogenous directional co-partial-moment matrices at degree 1.

This point is important. The NNS framework does not merely replace covariance with a new object; it decomposes covariance into its directional atoms. Classical covariance aggregates concordant and divergent directional dependence into one scalar entry. Co-partial moments preserve those directional components separately.

For the purpose of this note, no additional aggregate matrix formed only from CoLPM and CoUPM is required. The essential result is the full signed decomposition: covariance is recovered only when the concordant components and divergent components are combined with their appropriate signs.

Thus the central identity is:

$$\text{covariance} = \text{concordant directional dependence} - \text{divergent directional dependence}.$$

This is the cleanest bridge back to mean-variance optimization. The classical covariance matrix is itself an exogenous degree-1 directional composite. The NNS contribution is that it exposes the component matrices before they are collapsed into covariance.

9. Estrada as a Special Case Within the Directional Framework

Estrada's exogenous semicovariance matrix is essentially a special case of a lower concordant co-partial-moment matrix focused only on downside deviations. When the benchmark is $B = 0$, Estrada's matrix has entries of the form

$$\frac{1}{T} \sum_{t=1}^T \min(R_{i,t} - B, 0) \min(R_{j,t} - B, 0).$$

Since

$$\min(R_{i,t} - B, 0) = -\max(B - R_{i,t}, 0),$$

the product of two such terms is

$$\min(R_{i,t} - B, 0) \min(R_{j,t} - B, 0) = \max(B - R_{i,t}, 0) \max(B - R_{j,t}, 0).$$

Thus Estrada's matrix is equivalent, up to sign convention, to a concordant lower partial-moment construction using asset-level shortfalls.

Explicitly, for $n = 1$ and benchmark $B = 0$, Estrada's exogenous semicovariance entries are exactly the degree-1 concordant lower co-partial moment entries:

$$\Sigma_{ij0}^{\text{Estrada}} = \frac{1}{T} \sum_{t=1}^T \max(0 - R_{i,t}, 0) \max(0 - R_{j,t}, 0) = \text{CoLPM}_{ij}^{(1)}(0).$$

This is the bridge between Estrada's downside-only construction and the broader degree-1 covariance decomposition above: Estrada keeps the lower concordant component, while the full covariance reconstruction also includes upper concordance and both divergent components.

The hierarchy is therefore:

$$\text{Covariance} = \text{degree-1 signed combination of directional co-partial moments},$$

Estrada semicovariance = downside-only exogenous co-lower partial moment construction,

NNS co-partial-moment framework = full directional matrix algebra across concordant and divergent regions.

The degree-1 case is especially fundamental because it shows that the directional framework contains classical covariance as a special signed aggregation, while also exposing the directional pieces that covariance collapses.

11. Did Viole–Nawrocki Solve Markowitz’s Endogeneity Problem?

Yes, but in a precise sense. As established in Sections 3 and 5, the traditional cosemivariance matrix $S_{ijB}(\mathbf{x})$ is endogenous because its indicator $\mathbf{1}_{\{R_{p,t}(\mathbf{x}) < B\}}$ depends on the portfolio weights, while Estrada restores tractability by moving the downside condition to the asset level, yielding the symmetric exogenous matrix $\Sigma_B = \text{CoLPM}^{(1)}(B)$ (as shown in Section 10). Viole–Nawrocki go further still: rather than approximating the portfolio-conditioned object, they replace it entirely with a family of pairwise, benchmark-relative directional co-partial moment matrices — exogenous by construction, Gram-structured, and modular across downside, upside, concordant, and divergent dependence (Section 6). As the degree-1 decomposition in Section 9 shows, classical covariance itself is a signed aggregation of these same primitives, so the NNS framework subsumes mean-variance as a special case rather than competing with it.

11.2 Covariance as a Directional Composite

The degree-1 case is especially important because it shows that classical covariance itself is a signed combination of exogenous directional matrices.

With benchmark $h_i = \mu_i$, write

$$X_i - \mu_i = U_i - L_i,$$

where

$$U_i = \max(X_i - \mu_i, 0), \quad L_i = \max(\mu_i - X_i, 0).$$

Then

$$\text{Cov}(X_i, X_j) = E[(U_i - L_i)(U_j - L_j)].$$

Expanding,

$$\text{Cov}(X_i, X_j) = E[U_i U_j] + E[L_i L_j] - E[L_i U_j] - E[U_i L_j].$$

Therefore,

$$\Sigma = \text{CoLPM}^{(1)} + \text{CoUPM}^{(1)} - \text{CoDLPM}^{(1)} - \text{CoDUPM}^{(1)}$$

This means the classical covariance matrix is not external to the NNS framework. It is a signed aggregation of degree-1 exogenous directional co-partial moment matrices.

Thus the NNS framework does not merely repair semivariance. It supplies the atomic directional components from which covariance, downside risk, upside risk, and asymmetric dependence measures can be constructed.

11.3 What Is Solved, and What Is Not

The exact Markowitz portfolio semivariance remains endogenous:

$$\frac{1}{T} \sum_{t=1}^T [\min(R_{p,t}(\mathbf{x}) - B, 0)]^2$$

still depends on the portfolio weights through $R_{p,t}(\mathbf{x})$. Any matrix whose entries are conditioned on

$$R_{p,t}(\mathbf{x}) < B$$

will remain endogenous.

The NNS solution is therefore not to transform the original endogenous object into an exogenous one. The solution is to redefine the risk primitive so that the matrix is exogenous by construction.

In this sense, Viole–Nawrocki fully resolve the computational endogeneity problem. They eliminate the need to optimize over a matrix whose entries change with portfolio weights. The optimizer instead receives fixed directional matrices computed pairwise from asset returns and benchmarks.

11.4 Final Interpretation

Thus, Viole–Nawrocki do not merely patch the semivariance problem. They generalize the risk matrix itself (see Section 10 for the full conceptual hierarchy).

The endogeneity problem is eliminated by design: portfolio risk is no longer built from portfolio-conditioned downside states, but from exogenous directional dependence primitives. Classical covariance becomes one special signed aggregation of these primitives, while downside and upside risk become alternative benchmark-relative assemblies of the same underlying directional matrix system.

12. Summary Table

Matrix type	Downside or directional event defined by	Depends on $\mathbf{x}$?	Exogenous?	Symmetric?	PSD structure	Optimization consequence
Classical covariance	Central deviations of individual assets	No	Yes	Yes	Gram matrix (full-sample)	Standard mean-variance QP
Traditional cosemi-variance	Portfolio underperformance $R_p(\mathbf{x}) < B$	Yes	No	Not generally	Not guaranteed	Requires iterative or special numerical methods
Estrada semico-variance heuristic	Individual asset underperformance $R_i < B$	No	Yes	Yes	Gram-type downside matrix	Mean-semivariance can be treated like mean-variance
NNS co-partial moment matrices	Pairwise directional deviations relative to h	No	Yes	Yes	Gram-type directional matrices	Quadratic optimization with directional risk primitives

13. Main Result

The endogenous/exogenous distinction can be summarized as follows:

A downside risk matrix is endogenous if downside states are defined by $R_p(\mathbf{x})$.

A downside or directional risk matrix is exogenous if its entries are defined pairwise from asset returns and a fixed benchmark.

Traditional mean-semivariance optimization is difficult because the downside region changes with the portfolio weights. Estrada restores tractability by replacing portfolio-conditioned downside states with asset-conditioned downside states. The Viole–Nawrocki/NNS framework generalizes

this idea by defining risk and dependence through lower, upper, concordant, and divergent co-partial moments, all computed exogenously from benchmark-relative directional deviations.

14. Conclusion

The covariance matrix is exogenous because it is computed once from asset returns and remains fixed during optimization. Traditional cosemivariance is endogenous because it requires knowing which periods the **portfolio** underperforms the benchmark, and that classification depends on the very weights being solved for.

Estrada’s heuristic solves this computational problem by redefining semicovariance using individual asset downside deviations, producing a symmetric exogenous matrix usable in the same quadratic form as covariance.

The Viole–Nawrocki/NNS approach extends the solution: it treats directional partial moments and co-partial moments as the primitive objects, yielding exogenous, symmetric, positive semidefinite matrices that preserve asymmetry, nonlinear dependence, and tail-sensitive structure.

The progression is:

Variance $\rightarrow$ Endogenous Semivariance $\rightarrow$ Exogenous Semicovariance
 $\rightarrow$ Exogenous Directional Co-Partial Moment Matrices.

The NNS/Viole–Nawrocki contribution is therefore not that the original Markowitz semivariance matrix becomes exogenous, but that the need for such an endogenous matrix disappears. Directional co-partial moment matrices provide the exogenous primitives from which covariance, downside risk, upside risk, and asymmetric dependence can all be assembled.

In short: the endogeneity problem is eliminated by design, not cured by approximation.

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